

Supplementing language model-based mRNA translation efficiency prediction with structural priors

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1 Introduction

mRNA vaccines and therapeutics have become a major success of modern biomedicine, most visibly through the rapid development and high efficacy of the first COVID-19 mRNA vaccines [1]. mRNA vaccines offer favorable safety profiles because they do not carry the risk of insertional mutagenesis associated with DNA-based platforms [2]. They also elicit strong cellular immunity, including early mobilization of CD8+ T-cell responses [3]. More broadly, mRNA platforms are attractive biotechnology tools because RNA design is both rapid and modular, making the technology adaptable to diverse targets. One central challenge for mRNA therapeutics is controlling how efficiently a given mRNA is translated into protein, as translation efficiency (TE) directly shapes the functional output of the drug [2]. Translation initiation, the rate-limiting step, is governed by features in the 5' untranslated region (UTR) such as the Kozak sequence in mammals and the number of upstream open reading frames (uORFs) [4]. However, current evidence increasingly highlights the importance of the early coding sequence (CDS) context in regulating translation, notably through secondary structure elements that may reduce accessibility and hinder initiation [5] and corresponding abundance of tRNAs [6]. This creates a novel opportunity for therapeutic mRNA sequence optimization, as a recent study showed that CDS optimization of a COVID-19 mRNA vaccine elicited 128 times higher antibody titers in mice [7]. In sum, accurate measurement and prediction of mRNA translation efficiency essential for rational sequence design.

Current experimental approaches to measuring mRNA TE differ both in their readouts and in the biological contexts they assay. Polysome profiling estimates mean ribosome load (MRL) by separating mRNAs on a sucrose gradient according to the number of associated ribosomes [8]. This strategy underlies massively parallel reporter assays (MRPAs), in which synthetic constructs typically place a variable 5' UTR upstream of a fixed reporter coding sequence such as GFP. These assays provide highly reproducible measurements, with R^2 values reaching 0.95 [8], and have generated large-scale datasets for learning translation-relevant features encoded in 5' UTRs. Accordingly, models trained on such data, including FramePool [9] and Optimus 5-Prime [8], achieve strong predictive performance on synthetic benchmarks. However, this accuracy does not generalize well to endogenous tran-

scripts [10]. This suggests not only that the sequence determinants of translation differ between synthetic and endogenous 5' UTRs, but also that reporter-based datasets with a fixed coding sequence fail to capture 5' UTR-CDS interactions that are important for translational control in vivo [10].

To address the need for datasets that capture translational regulation in endogenous mRNAs, Zheng et al. assembled a large compendium of TE measurements comprising 1,282 human and 995 mouse samples [11]. Using this dataset, RiboNN, a convolutional neural network (CNN), achieved an R^2 of 0.62 for predicting mean TE across cell types and recovered known determinants of translational control, including uAUGs in the 5' UTR and optimal codons in the CDS [11]. However, CNN architectures may be suboptimal for this task because they favor local sequence patterns and may limit the capture of longer-range dependencies, including secondary structure-mediated interactions.

Recent RNA language models have shown promise across a broad range of RNA-related tasks, including ncRNA family classification and splice-site prediction [12]. However, a major limitation of transformer-based architectures is their quadratic memory complexity with sequence length. This has led most RNA LLMs to focus either on non-coding RNAs (ERNIE-RNA [13]) or on restricted mRNA segments, such as the 5' UTR (UTR-LM [14]). As a result, these models are unable to capture the interplay between the 5' UTR and the CDS. mRNABERT was designed to address this gap through three innovations: a dual-tokenization scheme that represents UTRs at nucleotide resolution and CDSs at codon resolution, FlashAttention to reduce memory traffic and improve training efficiency, and Attention with Linear Biases (ALiBi) to enable extrapolation to sequences longer than those seen during training [15]. On downstream benchmarks, mRNABERT improved performance across multiple mRNA design tasks and achieved a mean TE R^2 of 0.669 after fine-tuning on the RiboNN task at a sequence maximum length of 3066 tokens.

Because no fine-tuned weights were provided by the authors, this project first aimed to fine-tune the base mRNABERT model on the RiboNN task and reproduce its reported performance for internal use within the Laboratory for Biomolecular Modeling. Second, it sought to investigate the learnt TE-relevant features responsible for mRNABERT's improved predictive performance. Third, and most importantly, it examined whether incorporating RNA secondary struc-

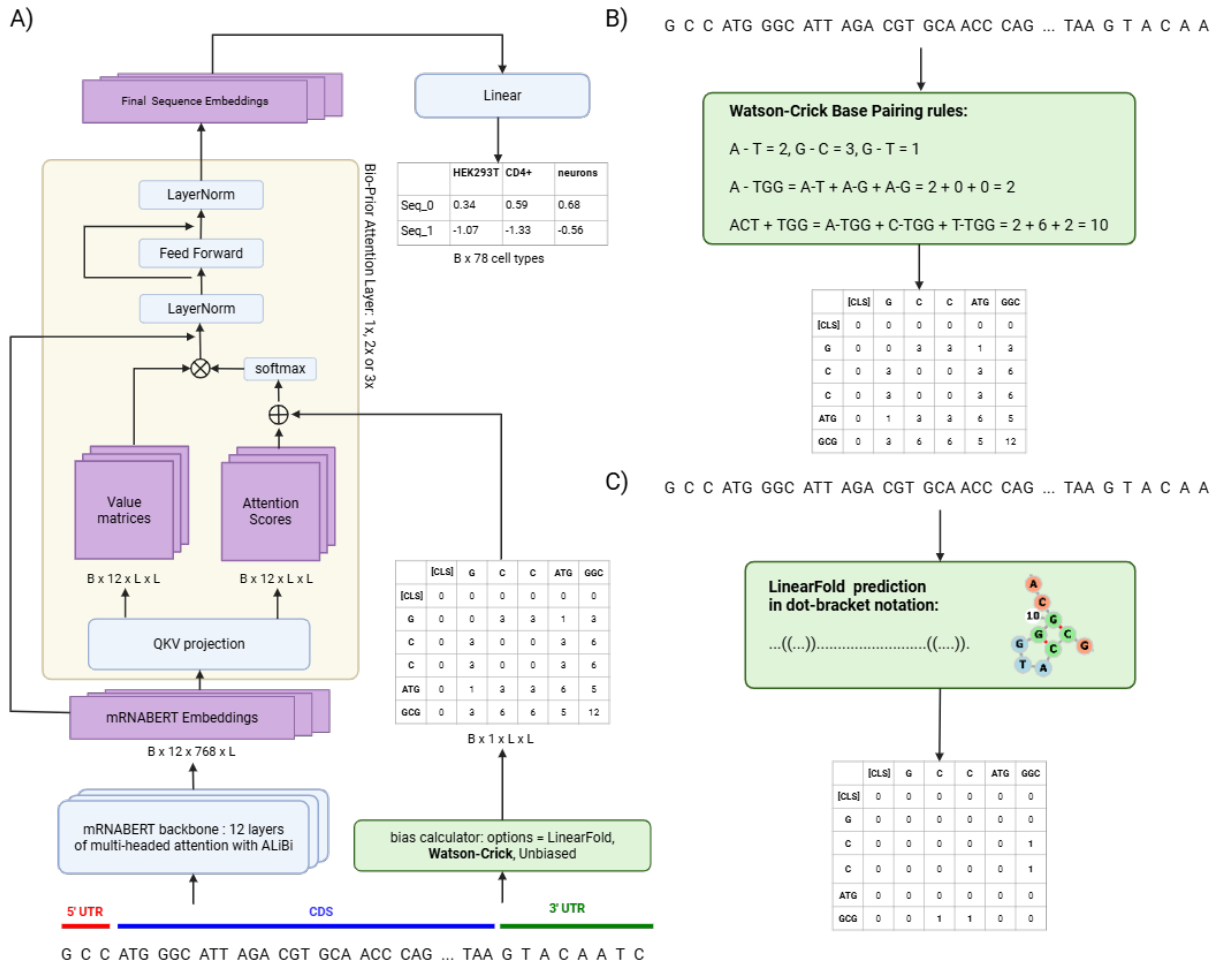


Figure 1: Model architecture and structural biasing schemes. A) Overview of the biased-attention model. The frozen mRNABERT backbone produces contextual embeddings, which are passed through additional multi-head attention layers whose scores can be modulated by an external bias matrix. Transcription efficiency values are predicted for 78 cell-types from final sequence representations through a linear layer. B) Watson-Crick-based biasing scheme. A bias matrix is constructed from canonical base-pairing rules, assigning scores of 3 to G-C=3, A-T, and 1 to G-T pairs. For two tokens, pairwise scores across all bases are summed to produce the final bias term. C) LinearFold-based biasing scheme. The input sequence is first folded with LinearFold, and the resulting dot-bracket structure is converted into a token-token bias matrix.

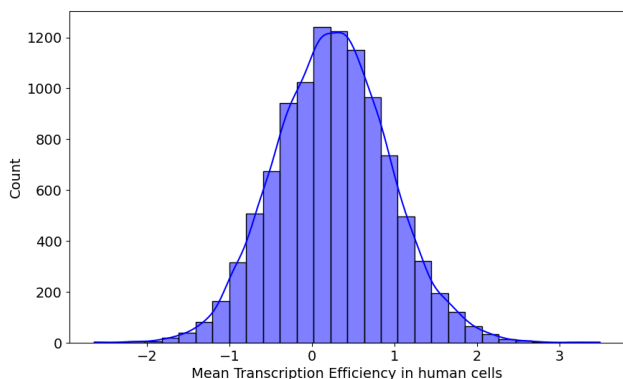


Figure 2: Distribution of mean transcription efficiency for the human RiboNN dataset.

ture information could further improve predictive accuracy. To this end, we introduced a biased-attention framework (Fig. 1A) in which the mRNABERT backbone remains frozen while standard multi-head attention layers are appended and biased by structural priors. We evaluated two biasing strategies: one based on standard Watson-Crick base-pairing and one based on LinearFold-predicted secondary structure.

2 Methods

2.1 Data and Benchmark Task

Zheng et al.’s datasets of translation efficiency (TE) measurements are comprised of 1,282 human and 995 mouse samples [11]. These measurements were derived from matched ribosome profiling (Ribo-seq) and RNA-seq data. Using a compositional regression approach, the authors converted these raw counts into cell-type-specific TE values, and most cell types showed substantial correlation with one another [11]. In the human dataset, the mean TE across cell types was 0.25 with a standard deviation of 0.69 (Fig. 2). The main performance metric throughout this study is R^2 between observed and predicted mean translation efficiency across cell types, and all analyses focused on the human dataset.

2.2 Model Architecture and Training

The model for predicting translation efficiency was built on top of a frozen mRNABERT backbone. The backbone consists of 12 multi-head bidirectional encoder layers with a hidden dimension of 768. mRNABERT was pre-trained on 18 million mRNA sequences, cropped at a maximum sequence length of 1024 tokens, using masked language modeling. mRNABERT was further tuned with contrastive learning to align CDS embeddings with those of their corresponding protein sequences [15].

The prediction head consisted of 1, 2, or 3 additional encoder layers similar to those of mRNABERT, each with a hidden dimension of 768, a feed-forward dimension of 3,072, and 8 attention heads, followed by a lin-

ear layer mapping the hidden representation to the 78 cell-type-specific output values (Fig. 1A). These encoder layers support an $L \times L$ bias term that can be injected to modulate attention patterns. We used this mechanism to incorporate secondary structure information into the model. Two biasing strategies were evaluated. The first assigns bias scores according to canonical Watson-Crick pairing, with scores of 3 for G-C, 2 for A-T, and 1 for G-T pairs (Fig. 1B). This strategy, used by ERNIE-RNA during pretraining [13], was shown to enable zero-shot recovery of RNA secondary structure from attention maps. The second passes sequences through the dynamic programming-based secondary structure predictor LinearFold [16] and converts its output directly into a token-token bias term ranging from 1 to 3 (Fig. 1C).

All models were trained using the AdamW optimizer with a learning rate of 8×10^{-5} and a batch size of 16 for 20 epochs, with early stopping if mean TE R^2 on the validation set failed to improve by 0.001 for 4 consecutive epochs. Unless stated otherwise, all reported performances and results correspond to models trained with a maximum sequence length of 1024 tokens.

To evaluate the impact of different mRNA sequence regions on performance, an ablation study was conducted, fine-tuning the model on the CDS only, 5’ UTR only, UTR 5’ and CDS only, or the entire sequence. Maximum sequence length was also varied between 1024 and 2044 tokens to evaluate the effect of longer sequence context. For this experiment, whose results are shown in Section 3, an alternative training scheme was used because the standard setup was too memory-intensive and did not allow expansion of sequence length to 2044. In this second scheme, a thin linear layer was appended to the frozen mRNABERT backbone, and the entire backbone was fine-tuned. The same hyperparameters were used for this setup.

2.3 Model evaluation and Statistical Testing

Ten-fold cross-validation was used to compare model variants. Sequences were assigned to folds according to the pre-existing RiboNN fold labels provided in the paper’s metadata [11]. Model performance was evaluated using R^2 on the mean predicted TE, and pairwise comparisons were performed with a difference-of-means t-test. To account for the dependence introduced by evaluating models across overlapping cross-validation folds, p-values were corrected using the Nadeau-Bengio correction and further adjusted with the Bonferroni correction to control for multiple comparisons.

2.4 Feature learning investigations

To assess whether mRNABERT captured codon optimality, we generated codon-optimized and codon-diminished variants for 724 wild-type test sequences chosen to have a 5’ UTR shorter than 300 nt. For each amino acid, the codon-optimized version was constructed by replacing the original codon with the hu-

man codon of highest usage frequency, whereas the codon-diminished version was generated using the least optimal human codon, as defined by a codon usage database [11]. This yielded paired sequence variants with altered codon adaptation index (CAI) while preserving the encoded protein. A one-layer model fine-tuned according to the first training setup, with no structural bias injected, was then used to score the wild-type and variant sequences, and the change in predicted translation efficiency was quantified as $\Delta\text{TE} = \text{TE}(\text{variant}) - \text{TE}(\text{wildtype})$. CAI distributions were computed for the wild-type and mutated sequences to verify the expected shift in codon optimality. To test whether the observed ΔTE values deviated significantly from zero, we applied a Wilcoxon signed-rank test for each of the two variant types.

To assess whether mRNABERT captured the effect of upstream AUGs, we performed AUG-insertion experiments on a one-layer model fine-tuned according to the first training setup, with no structural bias injected. In the first experiment, for each of 400 qualifying test-set transcripts (5'UTR > 501 nt, CDS > 300 nt), a single AUG motif was inserted at every admissible position within a window around the annotated start codon: out-of-frame positions 500 nt upstream through the 5'UTR, and in-frame positions every 3 nt from the start codon through 100 nt into the CDS. Each variant, along with the unmodified wild-type sequence, was scored with the fine-tuned model, and the change in predicted translation efficiency was quantified as $\Delta\text{TE} = \text{TE}(\text{variant}) - \text{TE}(\text{wildtype})$ for each insertion position, then averaged across transcripts. In the second experiment, to test the cumulative effect of upstream AUG burden independent of position, we inserted between 1 and 20 additional AUG motifs at random nucleotide positions 50–300 nt upstream of the start codon, and again computed ΔTE relative to each transcript's unmodified baseline.

2.5 Structure-attention correlation

To quantify the relationship between model attention and LinearFold-predicted RNA secondary structure, we compared attention scores with LinearFold-derived contact maps using Spearman rank correlation. For each of 400 test sequences, all token pairs predicted by LinearFold to form contacts were treated as positive pairs, and for every positive pair we sampled a negative pair with no predicted contact while matching the token-to-token distance of the positive pair. This led to a total of 3,299,075 sampled pairs. Attention scores were then extracted for the selected token pairs at each layer, and Spearman correlation ρ was computed between attention and contact score (between 0 and 3) across the sampled pairs. Correlations were reported separately for each layer of a one-layer model fine-tuned with no structural bias injected.

3 Results & Discussion

mRNABERT's performance relies on the interplay of the 5' UTR and CDS.

Without structural priors, fine-tuning mRNABERT on the RiboNN task achieved state-of-the-art performance, with a mean TE of 0.603 when sequences were truncated to 1024 tokens and 0.668 at 2044 tokens, surpassing RiboNN's R^2 of 0.62 (Fig. 3). In our ablation study, removing specific mRNA regions revealed that both the CDS-only model and the 5'-UTR-only model performed substantially worse, with R^2 values of 0.555 and 0.471, respectively. Removing the 3' UTR had no significant effect on performance in either the 1024- or 2044-token model.

Although the 5' UTR is the canonical site of translation initiation, its standalone predictive value was lower than that of the CDS, underscoring the dominant contribution of coding sequence information. This observation also helps explain the limited transferability of models trained on MRPA datasets in which only the 5' UTR varies, such as Optimus 5-Prime [8]. That removing the 3'UTR had no effect in the 2044-token variant suggests that the performance gain from longer inputs arises mainly from better coverage of the distal CDS rather than from modeling 3' UTR interactions. The impact of the 3' UTR on translation efficiency in vivo remains a matter of debate. 3' UTR length, miRNA-complementary elements and RNA-binding protein (RBP) interfaces are known regulator of initiation and are shown to have an essential role in specific contexts, such as for proto-oncogenes [17]. However, a previous study reported that 3' UTR isoforms had only a modest effect on translation efficiency for most mRNAs tested [18]. RiboNN likewise assigned the 3' UTR a limited contribution of 5% [11]. The absence of a detectable 3' UTR contribution in mRNABERT may reflect the fact that 52.33% of sequences in the dataset still had to be truncated at 2044 tokens, limiting the model's ability to capture the 3' UTR length effects and distal regulatory elements that likely constitute the main mode of 3' UTR-mediated translational control.

mRNABERT learns optimal human codons

Since mRNABERT was shown to rely strongly on the CDS, we next asked whether it captures the effect of codon optimality on translation efficiency. Synonymous codons are translated at different rates depending on the abundance of cognate and near-cognate tRNAs, giving rise to codon bias in many genomes, particularly in highly expressed genes [19]. Ribosome profiling in yeast revealed an inverse relationship between tRNA abundance and codon dwell time [20], and antibody-based profiling extended this observation to mammalian cells [21]. One proposed mechanism is that slow decoding at non-optimal codons causes ribosome crowding near the start codon, which in turn reduces new initiation events at the rate-limiting step.

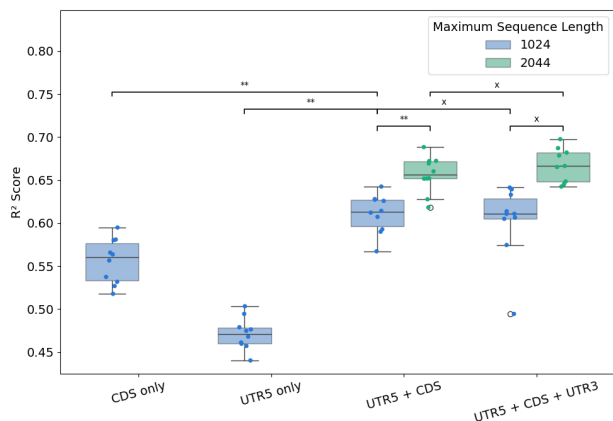


Figure 3: Impact of included mRNA sequence regions on model performance, measured by R^2 between observed and predicted mean TE. Model performance was assessed using 10-fold cross-validation. Boxplots are colored according to the model’s maximum input sequence length (1024 or 2044 tokens). Statistical comparisons were performed using mean-difference t-tests with both Nadeau-Bengio and Bonferroni corrections. ** indicates a corrected p -value below 0.01, whereas X denotes no statistically significant difference.

Codon-optimized variants all had codon adaptation index (CAI) of 1, as they used the optimal synonymous codon at each position. In contrast, codon-diminished variants used the least optimal codon for each amino acid, resulting in a mean CAI of 0.42 compared with 0.76 for wildtype (Fig. 4A). The average Δ TE was 0.75 for codon-optimized sequences, indicating a marked increase in predicted translation efficiency (Fig. 4B). By contrast, no significant difference was observed between wildtype and codon-diminished sequences.

These results may indicate that mRNABERT primarily learns the impact of highly optimal codons, but is less sensitive to gradations between intermediate and low CAI. Another possibility is that the model is capturing a threshold effect, whereby translation efficiency changes only once codon usage goes above a certain level of optimality. The similarly high standard deviations in Δ TE, 0.46 for codon-optimized sequences and 0.44 for codon-diminished sequences, further suggest that the effect of CAI on TE is strongly gene-dependent. More fine-grained analyses, such as stratifying Δ TE by sequence length or protein family, may help resolve these effects.

mRNABERT learns the impact of uAUGs

We investigated whether mRNABERT had learned a well-established repressor of translation in endogenous mRNAs: AUG codons upstream of the canonical start codon. These codons can create uORFs that repress downstream translation by diverting scanning ribosomes away from the main coding sequence [22]. To test this, we performed insertional analyses of AUG motifs on 400 sequences. Within the 5’UTR, single AUG motif insertion produced modest changes in Δ TE, ranging from -0.019 to -0.007. One exception was a pronounced dip at position -2, where Δ TE reached -0.03. Because an AUG inserted at -2 re-

places the strongly conserved purine at position -4 of the Kozak sequence with a thymine, this local decrease likely reflects disruption of the initiation context, indicating that the model has also learned the importance of the Kozak sequence for translation initiation. In contrast, inserting AUG motifs within the CDS had no biologically meaningful effect, with Δ TE values remaining above -0.003.

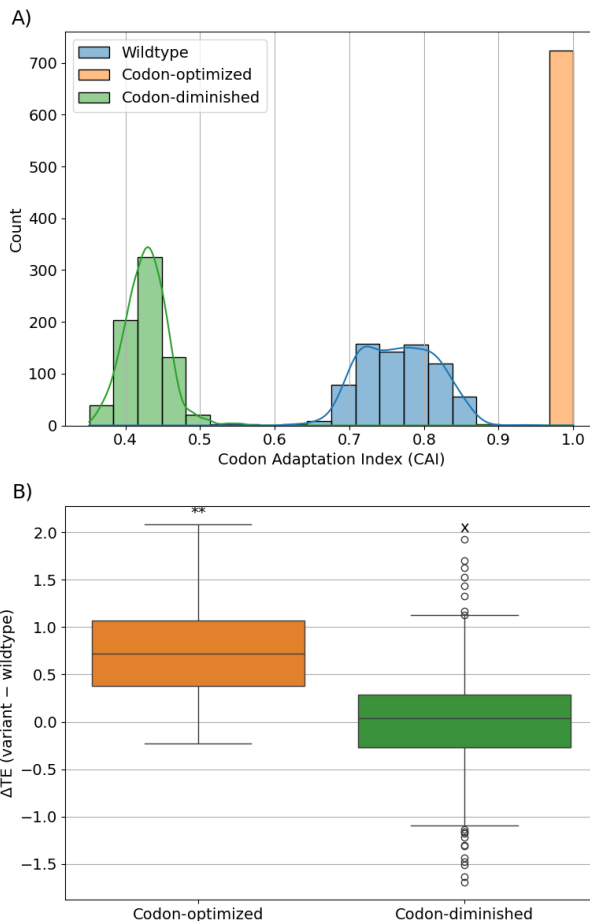


Figure 4: **mRNABERT predicts the impact of optimal codons on translation efficiency.** A) Distribution of codon adaptation index (CAI) values for wildtype, codon-optimized, and codon-diminished sequences. B) Model-predicted changes in translation efficiency, expressed as Δ TE (variant - wildtype), for codon-optimized and codon-diminished variants. ** denotes a mean Δ TE significantly different from 0 with $p < 0.01$ and X denotes no significant difference.

Multiple uAUGs can act synergistically to repress translation by stalling ribosomes as they scan for the start codon [22]. Hence, we next examined the effect of uAUG number on predicted mean TE. Increasing the number of inserted uAUGs produced a progressive decrease in mean Δ TE, from -0.016 for one inserted motif to -0.138 for 20 motifs (Fig. 5B), indicating that the model has learnt this compounding mechanism.

Biased models do not improve TE prediction

We tested whether adding secondary-structure-derived token-pair bias improves prediction by comparing a null-bias baseline against the two Bio-Prior variants

shown in Fig. 1. Across 10-fold cross-validation, all nine model configurations achieved similar mean TE R^2 values, ranging from 0.60 to 0.62 (Fig. 6). No pairwise comparison reached statistical significance. The three-layer Bio-Prior models showed a slight, non-significant decline in performance, consistent with the overfitting pattern visible in their loss curves (Fig. S1).

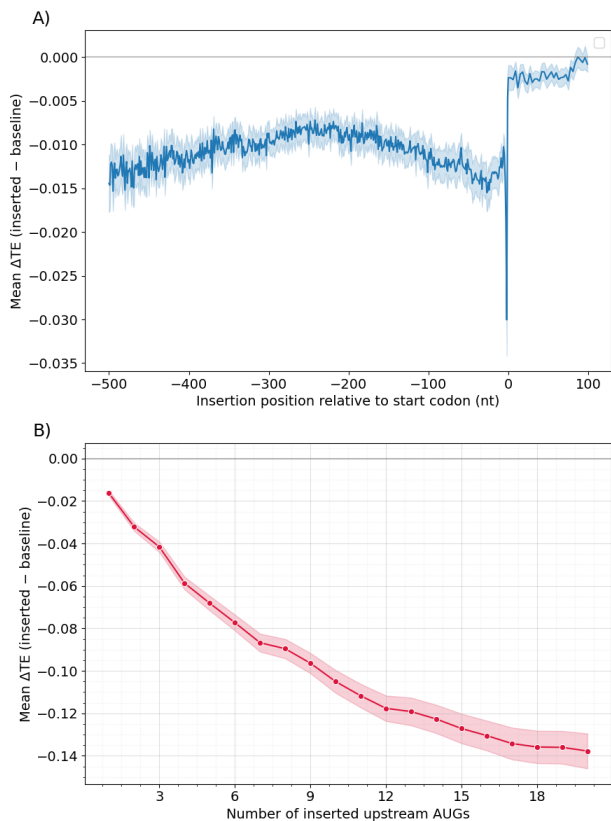


Figure 5: Insertional analysis of upstream AUG motifs and predicted translation efficiency. A) Mean model-predicted change in translation efficiency, expressed as ΔTE relative to the baseline sequence, as a function of the position of a single inserted AUG motif. Position 0 corresponds to the start codon. Insertions within the 5'UTR produce modest decreases in predicted TE, with a pronounced dip at position -2, whereas insertions within the CDS have little effect. B) Mean model-predicted ΔTE as a function of the number of inserted upstream AUG motifs. Increasing the number of uAUGs leads to a progressive decrease in predicted translation efficiency, consistent with cooperative repression by multiple uORFs. Shaded regions indicate variability across test sequences.

Secondary structure emerges from fine-tuning on RiboNN without explicit bias injection

It is well established that mRNA secondary structure influences translation efficiency, although studies disagree on which regions are most important and on the magnitude of the effect [23], [24]. We therefore asked why adding secondary-structure information did not improve predictive accuracy. We considered two hypotheses: first, that the mRNABERT backbone had already learned an internal representation of secondary structure through masked language modeling, despite never being trained on structural annotations; second, that the additional attention layers we introduced

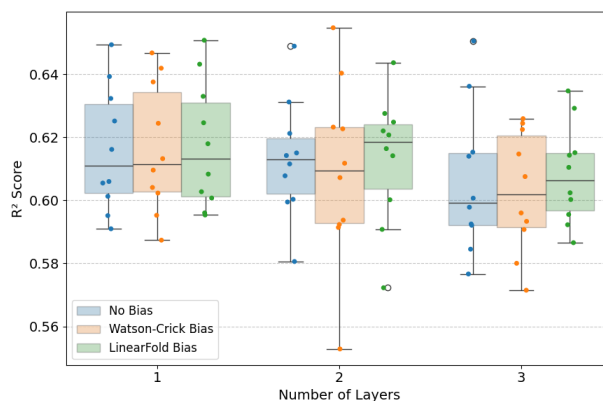


Figure 6: Fine-tuned mRNABERT models without bias injection learn to attend to secondary structure element. A) LinearFold-predicted secondary structure of one transcript from human gene FAM221A. Blue nucleic acids correspond to hairpin loops, green to stems, yellow to interior loops, red to multiloops and orange to unpaired regions. B) Corresponding attention heatmap for the same sequence, as predicted by an unbiased one-layer mRNABERT model following the architecture in Fig. 1. Attention scores are shown on a log scale. Red marks indicate token pairs predicted to form contacts by LinearFold. The lower annotation shows the token layout across the 5'UTR, CDS, and 3'UTR. A-B) Dashed black and pink boxes highlight two stem regions that the model attends to.

could learn structural features relevant to TE prediction even without explicit LinearFold or Watson-Crick biases. In either case, the explicit bias would be redundant, because the model would already be capturing the relevant structural information through its own learned representations.

To test these hypotheses, we examined whether attention scores from the backbone and Bio-Prior layers correlated with the contact maps predicted by LinearFold. The 12 layers of the mRNABERT backbone all showed only minimal correlation between LinearFold contacts and attention scores, with ρ values ranging from 0.042 to 0.065. In the one-layer unbiased model, the attention head trained for TE prediction showed roughly a threefold increase in correlation, reaching $\rho = 0.18$. Although still modest, this increase suggests that the fine-tuned heads learn some secondary-structure features relevant to TE prediction, providing support for the second hypothesis. The model may, for example, focus on secondary structure in functionally important regions such as the vicinity of the start codon, but further analysis is needed to determine which features it captures.

Visualization of the attention heatmaps first reveals that the model attributes significantly higher attention scores to the CDS region compared to the UTRs (Fig. 7B). In the example of transcript ENST00000344962.9, corresponding to human gene FAM221A, the model seems to attend strongly to two stem loops, highlighted by black and pink dashed rectangles (Fig. 7A), indicating that these structures may be affecting TE.

4 Conclusion

In this project, we fine-tuned mRNABERT on the Ri-boNN task and found that the model captures important determinants of endogenous translation efficiency, with performance improving when longer sequence contexts were available. The ablation analyses showed that the CDS contributed more predictive information than the 5'UTR alone, while removing the 3'UTR had little effect in this setup. Further perturbation experiments indicated that the model learned biologically meaningful signals related to codon optimality and upstream AUG motifs, since codon-optimized sequences produced a strong increase in predicted translation efficiency and uAUG insertion progressively reduced predicted translation efficiency. We also attempted to supplement the model with secondary-structure information through an attention-biasing approach, but found no improvement in performance. Preliminary analyses suggest that this may be because mRNABERT already learns relevant structural features when fine-tuned for TE prediction. In terms of performance, only increasing the maximum accommodated sequence length during fine-tuning improved mean TE R^2 .

Several limitations should be kept in mind when interpreting these findings. First, the structural biases relied on LinearFold predictions as though they accurately reflected the true RNA secondary structure, although LinearFold is an approximate algorithm that cannot resolve structures such as pseudoknots [16]. Experimentally resolved structures or more accurate methods may therefore provide an advantage over LinearFold-based biases. Additionally, further engineering efforts could be directed toward improving the structural prior, for example by tuning or learning the bias magnitude, or by integrating structural information through other approaches.

Second, some analyses would benefit from finer-grained controls. In the uAUG experiment, the impact of inserted AUG motifs on translation efficiency is known to depend strongly on initiation context and reading-frame overlap [22], both of which were not systematically isolated. In the codon optimality experiment, the observed changes in predicted translation efficiency may also reflect confounders such as altered GC content and changes in mRNA secondary structure, rather than codon usage alone. Future work should therefore disentangle these factors with more controlled perturbations.

Finally, an important limitation is that the model does not represent chemically modified or non-canonical nucleotides, such as N1-methylpseudouridine, even though such modifications are central to modern mRNA vaccine design and can strongly affect translation, stability, and immunogenicity [25]. The model also has a very high number of parameters, whereas simpler machine-learning approaches were found to achieve only slightly lower performance [11].

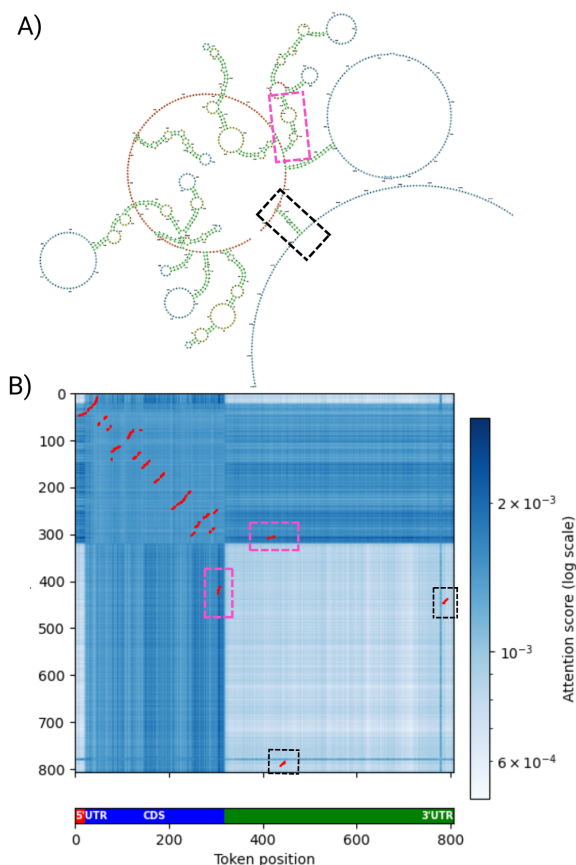


Figure 7: Fine-tuned mRNABERT models without bias injection learn to attend to secondary structure element.

A) LinearFold-predicted secondary structure of one transcript from human gene FAM221A. Blue nucleic acids correspond to hairpin loops, green to stems, yellow to interior loops, red to multiloops and orange to unpaired regions. B) Corresponding attention heatmap for the same sequence, as predicted by an unbiased one-layer mRNABERT model following the architecture in Fig. 1. Attention scores are shown on a log scale. Red marks indicate token pairs predicted to form contacts by LinearFold. The lower annotation shows the token layout across the 5'UTR, CDS, and 3'UTR. A-B) Dashed black and pink boxes highlight two stem regions that the model attends to.

Overall, the project supports the use of specialized mRNA language models to predict therapeutically relevant features such as translation efficiency and paves the way for their application in the optimization of mRNA vaccines and therapeutics. In practice, translation efficiency must be balanced against other sometimes competing objectives, including mRNA stability, for optimal therapeutic design.

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References

- [1] M. W. Tenforde et al., “Effectiveness of Severe Acute Respiratory Syndrome Coronavirus 2 Messenger RNA Vaccines for Preventing Coronavirus Disease 2019 Hospitalizations in the United States,” *Clinical Infectious Diseases*, vol. 74, no. 9, pp. 1515–1524, May 2022.
- [2] S. Guan and J. Rosenecker, “Nanotechnologies in delivery of mRNA therapeutics using nonviral vector-based delivery systems,” *Gene Therapy*, vol. 24, no. 3, pp. 133–143, Mar. 2017.
- [3] V. Oberhardt et al., “Rapid and stable mobilization of CD8+ T cells by SARS-CoV-2 mRNA vaccine,” *Nature*, vol. 597, no. 7875, pp. 268–273, Sep. 2021.
- [4] A. Simonetti, E. Guca, A. Bochler, L. Kuhn, and Y. Hashem, “Structural Insights into the Mammalian Late-Stage Initiation Complexes,” *Cell Reports*, vol. 31, no. 1, p. 107497, Apr. 2020.
- [5] S. Höllner and M. Jeschek, “Ultradeep characterisation of translational sequence determinants refutes rare-codon hypothesis and unveils quadruplet base pairing of initiator tRNA and transcript,” *Nucleic Acids Research*, vol. 51, no. 5, pp. 2377–2396, Mar. 2023.
- [6] M. Torrent, G. Chalancon, N. S. de Groot, A. Wuster, and M. Madan Babu, “Cells alter their tRNA abundance to selectively regulate protein synthesis during stress conditions,” *Science Signaling*, vol. 11, no. 546, eaat6409, Sep. 2018.
- [7] H. Zhang et al., “Algorithm for optimized mRNA design improves stability and immunogenicity,” *Nature*, vol. 621, no. 7978, pp. 396–403, Sep. 2023.
- [8] P. J. Sample et al., “Human 5 UTR design and variant effect prediction from a massively parallel translation assay,” *Nature Biotechnology*, vol. 37, no. 7, pp. 803–809, Jul. 2019.
- [9] A. Karollus, Ž. Avsec, and J. Gagneur, “Predicting mean ribosome load for 5’UTR of any length using deep learning,” *PLOS Computational Biology*, vol. 17, no. 5, e1008982, May 2021.
- [10] N. Schlusser, A. González, M. Pandey, and M. Zavolan, “Current limitations in predicting mRNA translation with deep learning models,” *Genome Biology*, vol. 25, no. 1, p. 227, Aug. 2024.
- [11] D. Zheng et al., “Predicting the translation efficiency of messenger RNA in mammalian cells,” *Nature Biotechnology*, vol. 44, no. 5, pp. 783–796, May 2026.
- [12] Y. Ren et al., *BEACON: Benchmark for Comprehensive RNA Tasks and Language Models*, arXiv:2406.10391 [q-bio.QM], Dec. 2024.
- [13] W. Yin et al., “ERNIE-RNA: An RNA language model with structure-enhanced representations,” *Nature Communications*, vol. 16, no. 1, p. 10076, Nov. 2025.
- [14] Y. Chu et al., “A 5 UTR language model for decoding untranslated regions of mRNA and function predictions,” *Nature Machine Intelligence*, vol. 6, no. 4, pp. 449–460, Apr. 2024.
- [15] Y. Xiong, A. Wang, Y. Kang, C. Shen, C.-Y. Hsieh, and T. Hou, “mRNABERT: Advancing mRNA sequence design with a universal language model and comprehensive dataset,” *Nature Communications*, vol. 16, no. 1, p. 10371, Nov. 2025.
- [16] L. Huang et al., “LinearFold: Linear-time approximate RNA folding by 5’-to-3’ dynamic programming and beam search,” *Bioinformatics*, vol. 35, no. 14, pp. i295–i304, Jul. 2019.
- [17] C. Mayr and D. P. Bartel, “Widespread shortening of 3’UTRs by alternative cleavage and polyadenylation activates oncogenes in cancer cells,” *Cell*, vol. 138, no. 4, pp. 673–684, Aug. 2009.
- [18] N. Spies, C. B. Burge, and D. P. Bartel, “3’ UTR-isoform choice has limited influence on the stability and translational efficiency of most mRNAs in mouse fibroblasts,” *Genome Research*, vol. 23, no. 12, pp. 2078–2090, Dec. 2013.
- [19] G. Hanson and J. Collier, “Codon optimality, bias and usage in translation and mRNA decay,” *Nature reviews. Molecular cell biology*, vol. 19, no. 1, pp. 20–30, Jan. 2018.
- [20] J. A. Hussmann, S. Patchett, A. Johnson, S. Sawyer, and W. H. Press, “Understanding Biases in Ribosome Profiling Experiments Reveals Signatures of Translation Dynamics in Yeast,” *PLOS Genetics*, vol. 11, no. 12, e1005732, Dec. 2015.
- [21] X. Yan, T. A. Hoek, R. D. Vale, and M. E. Tanenbaum, “Dynamics of Translation of Single mRNA Molecules In Vivo,” *Cell*, vol. 165, no. 4, pp. 976–989, May 2016.
- [22] T. G. Johnstone, A. A. Bazzini, and A. J. Giraldez, “Upstream ORFs are prevalent translational repressors in vertebrates,” *The EMBO Journal*, vol. 35, no. 7, pp. 706–723, Apr. 2016.
- [23] J. Lin, Y. Chen, Y. Zhang, H. Lin, and Z. Ouyang, “Deciphering the role of RNA structure in translation efficiency,” *BMC Bioinformatics*, vol. 23, no. Suppl 3, p. 559, Dec. 2022.
- [24] T. Tuller, Y. Y. Waldman, M. Kupiec, and E. Ruppin, “Translation efficiency is determined by both codon bias and folding energy,” *Proceedings of the National Academy of Sciences*, vol. 107, no. 8, pp. 3645–3650, Feb. 2010.
- [25] N. Pardi et al., “Nucleoside-modified mRNA vaccines induce potent T follicular helper and germinal center B cell responses,” *The Journal of Experimental Medicine*, vol. 215, no. 6, pp. 1571–1588, Jun. 2018.

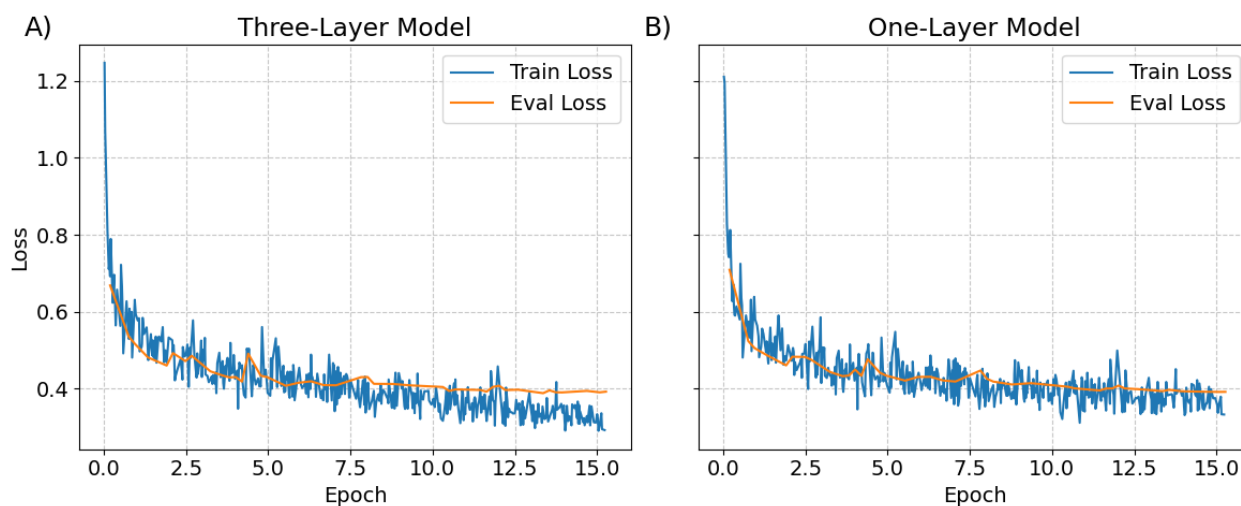


Figure S1: **Training and validation loss curves for representative Bio-Prior models.** A) Training loss and evaluation loss across epochs for a three-layer model. Both curves decrease rapidly during the first few epochs and then plateau, with the evaluation loss flattening earlier than the training loss, suggesting mild overfitting at later epochs. B) Training loss and evaluation loss across epochs for a one-layer Bio-Prior model. Loss decreases in a similar manner, but the evaluation curve remains more stable and closely tracks the training curve.